从 Snapshot 副本还原文件

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从 Snapshot 副本还原文件

从 NFS 或 SMB 客户端上的 Snapshot 副本还原文件

NFS 或 SMB 客户端上的用户可以直接从 Snapshot 副本还原文件，而无需存储系统管理员干预。

文件系统中的每个目录都包含一个名为 `.snapshot` 的子目录， NFS 和 SMB 用户可以访问该子目录。`.snapshot` 子目录包含与卷的 Snapshot 副本对应的子目录：

```
$ ls .snapshot
daily.2017-05-14_0013/              hourly.2017-05-15_1106/
daily.2017-05-15_0012/              hourly.2017-05-15_1206/
hourly.2017-05-15_1006/             hourly.2017-05-15_1306/
```

每个子目录都包含 Snapshot 副本引用的文件。如果用户意外删除或覆盖某个文件，他们可以将该文件从 Snapshot 子目录复制到读写目录，从而将该文件还原到父读写目录：

```
$ ls my.txt
ls: my.txt: No such file or directory
$ ls .snapshot
daily.2017-05-14_0013/              hourly.2017-05-15_1106/
daily.2017-05-15_0012/              hourly.2017-05-15_1206/
hourly.2017-05-15_1006/             hourly.2017-05-15_1306/
$ ls .snapshot/hourly.2017-05-15_1306/my.txt
my.txt
$ cp .snapshot/hourly.2017-05-15_1306/my.txt .
$ ls my.txt
my.txt
```

启用和禁用 NFS 和 SMB 客户端对 Snapshot 副本目录的访问

要确定 NFS 和 SMB 客户端是否可以查看 Snapshot 副本目录以从 Snapshot 副本还原文件或 LUN ，您可以使用 `volume modify` 命令的 `-snapdir-access` 选项启用和禁用对 Snapshot 副本目录的访问。

步骤

1. 检查 Snapshot 目录访问状态：

   ```
   `volume show -vserver _SVM_name_ -volume _vol_name_ -fields snapdir-
   access`
   ```

示例

```
clus1::> volume show -vserver vs0 -volume vol1 -fields snapdir-access
vserver volume snapdir-access
------- ------ --------------
vs0     vol1   false
```

2. 启用或禁用 Snapshot 副本目录访问：

   `volume modify -volume vol_name -snapdir-access true_false`

   以下示例将在 vol1 上启用 Snapshot 副本目录访问：

```
clus1::> volume modify -volume vol1 -snapdir-access true
Volume modify successful on volume vol1 of Vserver vs0.
```

从 Snapshot 副本还原单个文件

您可以使用 `volume snapshot restore-file` 命令从 Snapshot 副本还原单个文件或 LUN 。如果您不想替换现有文件，则可以将此文件还原到父读写卷中的其他位置。

如果要还原现有 LUN ，则会创建一个 LUN 克隆，并以 Snapshot 副本的形式进行备份。在还原操作期间，您可以读取和写入 LUN 。

默认情况下，包含流的文件会进行还原。

步骤

1. 列出卷中的 Snapshot 副本：

   `volume snapshot show -vserver svm -volume volum`

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `vol1` 中的 Snapshot 副本：

```
clus1::> volume snapshot show -vserver vs1 -volume vol1

Vserver Volume Snapshot                 State    Size  Total% Used%
------- ------ ---------- ----------- ------   -----  ------ -----
vs1  vol1   hourly.2013-01-25_0005  valid   224KB     0%    0%
               daily.2013-01-25_0010   valid   92KB      0%    0%
               hourly.2013-01-25_0105  valid   228KB     0%    0%
               hourly.2013-01-25_0205  valid   236KB     0%    0%
               hourly.2013-01-25_0305  valid   244KB     0%    0%
               hourly.2013-01-25_0405  valid   244KB     0%    0%
               hourly.2013-01-25_0505  valid   244KB     0%    0%

7 entries were displayed.
```

2. 从 Snapshot 副本还原文件：

   `volume snapshot restore-file -vserver` *`svm`* `-volume` *`volume`*`-snapshot` *`snapshot`* `-path` *`file_path`* `-restore-path` *`destination_path`*

   有关完整的命令语法，请参见手册页。

   以下示例将还原文件 `myfile.txt`：

```
cluster1::> volume snapshot restore-file -vserver vs0 -volume vol1
-snapshot daily.2013-01-25_0010 -path /myfile.txt
```

从 Snapshot 副本还原文件的部分内容

您可以使用 `volume snapshot partial-restore-file` 命令将一系列数据从 Snapshot 副本还原到 LUN 或 NFS 或 SMB 容器文件，前提是您知道数据的起始字节偏移量和字节计数。您可以使用此命令还原在同一 LUN 中存储多个数据库的主机上的一个数据库。

步骤

1. 列出卷中的 Snapshot 副本：

   `volume snapshot show -vserver` *`svm`* `-volume` *`volum`*

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `vol1` 中的 Snapshot 副本：

```
clus1::> volume snapshot show -vserver vs1 -volume vol1

Vserver Volume Snapshot                 State    Size  Total% Used%
------- ------ ---------- ----------- ------   -----  ------ -----
vs1  vol1   hourly.2013-01-25_0005  valid   224KB     0%    0%
               daily.2013-01-25_0010   valid   92KB      0%    0%
               hourly.2013-01-25_0105  valid   228KB     0%    0%
               hourly.2013-01-25_0205  valid   236KB     0%    0%
               hourly.2013-01-25_0305  valid   244KB     0%    0%
               hourly.2013-01-25_0405  valid   244KB     0%    0%
               hourly.2013-01-25_0505  valid   244KB     0%    0%

7 entries were displayed.
```

2. 从 Snapshot 副本还原文件的部分内容：

   `volume snapshot partial-restore-file -vserver` *`svm`* `-volume` *`volume`*`-snapshot` *`snapshot`* `-path` *`file_path`* `-start-byte` *`staring_byte`* `-byte-count` `_byte_count_count`

   起始字节偏移和字节计数必须是 4 ， 096 的倍数。

   以下示例将还原文件 `myfile.txt` 的前 4 ， 096 个字节：

```
cluster1::> volume snapshot partial-restore-file -vserver vs0 -volume
vol1 -snapshot daily.2013-01-25_0010 -path /myfile.txt -start-byte 0
-byte-count 4096
```

从 Snapshot 副本还原卷的内容

您可以使用 `volume snapshot restore` 命令从 Snapshot 副本还原卷的内容。

如果卷具有 SnapMirror 关系，请在从 Snapshot 副本还原后立即手动复制卷的所有镜像副本。否则，可能会导致镜像副本不可用，必须删除并重新创建这些副本。

步骤

1. 列出卷中的 Snapshot 副本：

   `volume snapshot show -vserver` *`svm`* `-volume` *`volum`*

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `vol1` 中的 Snapshot 副本：

```
clus1::> volume snapshot show -vserver vs1 -volume vol1

Vserver Volume Snapshot                State    Size  Total% Used%
------- ------ ---------- ----------- ------   -----  ------ -----
vs1  vol1   hourly.2013-01-25_0005  valid   224KB     0%    0%
               daily.2013-01-25_0010   valid   92KB      0%    0%
               hourly.2013-01-25_0105  valid   228KB     0%    0%
               hourly.2013-01-25_0205  valid   236KB     0%    0%
               hourly.2013-01-25_0305  valid   244KB     0%    0%
               hourly.2013-01-25_0405  valid   244KB     0%    0%
               hourly.2013-01-25_0505  valid   244KB     0%    0%

7 entries were displayed.
```

2. 从 Snapshot 副本还原卷的内容：

   `volume snapshot restore -vserver` *`svm`* `-volume` *`volume" -snapshot _snapshot`*

   有关完整的命令语法，请参见手册页。

   以下示例将还原 `vol1` 的内容：

```
cluster1::> volume snapshot restore -vserver vs0 -volume vol1 -snapshot
daily.2013-01-25_0010
```

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